

STA322 - Project 1

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Introduction

```
library(survey)
```

```
Loading required package: grid
```

```
Loading required package: Matrix
```

```
Loading required package: survival
```

```
Attaching package: 'survey'
```

```
The following object is masked from 'package:graphics':
```

```
dotchart
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
x tidyr::expand() masks Matrix::expand()
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
x tidyr::pack() masks Matrix::pack()
x tidyr::unpack() masks Matrix::unpack()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(pps)
library(dplyr)
```

```
# load and reorganize data
```

```
trinity <- read_csv("trinity_faculty.csv")
```

```
Rows: 41 Columns: 5
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Dept name
```

```
dbl (4): Stratum #, # of faculty, Proportional allocation, Rounded allocation
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trinity <- trinity |>
  select(department = "Dept name", M_i = "# of faculty")

# certainty departments (pi_i = 1), same 7 as in the sampling step
certainty_depts <- c("Economics", "Physics", "Romance Studies", "Music",
  "Biology", "Mathematics", "Psychology & Neuroscience")

certainty <- trinity |>
  filter(department %in% certainty_depts) |>
  mutate(pi_i = 1)

# remaining departments: 13 more selected, pi_i = 13 * M_i / sum(M_i)
remaining <- trinity |>
  filter(!department %in% certainty_depts)
remaining$pi_i <- 13 * remaining$M_i / sum(remaining$M_i)

dept_info <- bind_rows(certainty, remaining)
```

```
# systematic PPS sampling
```

```
trinity <- read_csv("trinity_faculty.csv") |>
  select(stratum = "Stratum #",
         department = "Dept name",
         M_i = "# of faculty")
```

```
Rows: 41 Columns: 5
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Dept name
```

```
dbl (4): Stratum #, # of faculty, Proportional allocation, Rounded allocation
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# 2. define n_dept, compute t_x and the first-pass pi_i
```

```
n_dept <- 20 # number of departments to select
```

```
t_x <- sum(trinity$M_i) # total number of faculty across all departments
```

```
trinity$pi_i <- n_dept * trinity$M_i / t_x # first-order inclusion probability
```

```
max(trinity$pi_i) # should be 1.378, so some departments exceed 1
```

```
[1] 1.378151
```

```
# dealing with certainty cases
```

```
# separate initial certainty and non-certainty departments
```

```
certainty <- trinity |>
```

```
  filter(pi_i >= 1) |>
```

```
  mutate(pi_i = 1)
```

```
remaining <- trinity |>
```

```
  filter(pi_i < 1)
```

```
n_remaining <- n_dept - nrow(certainty) # calculate number of additional departments to select
```

```
remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i) # recalculate pi_i for remaining
```

```
# if recalculation creates additional certainty PSUs, repeat until all remaining probabilities are 1
while (any(remaining$pi_i >= 1)) {
  new_certainty <- remaining |>
    filter(pi_i >= 1) |>
    mutate(pi_i = 1)
  certainty <- bind_rows(certainty, new_certainty)
  remaining <- remaining |>
    filter(pi_i < 1)
  n_remaining <- n_dept - nrow(certainty)
  remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i)
}
```

```
# results check
certainty |>
  select(department, M_i, pi_i) # 7 certainty cases identified (1 additional throughout recalculation)
```

```
# A tibble: 7 x 3
  department      M_i pi_i
  <chr>          <dbl> <dbl>
1 Biology         77     1
2 Economics        81     1
3 Mathematics      82     1
4 Music           62     1
5 Physics          60     1
6 Romance Studies  66     1
7 Psychology & Neuroscience 59     1
```

```
nrow(certainty)
```

```
[1] 7
```

```
n_remaining
```

```
[1] 13
```

```
max(remaining$pi_i)
```

```
[1] 0.9246088
```

```

set.seed(919)
# select remaining cases with pps & add certainty cases

selected_depts <- ppss(remaining$M_i, n_remaining) # returns numbers of selected rows

pps_remaining <- remaining[selected_depts, ] # keep selected departments

pps_sample <- bind_rows(certainty, pps_remaining) # combine certainty psus with pps-selected

nrow(pps_sample) # check 20 departments were selected

```

```
[1] 20
```

```

# view selected departments
pps_sample |>
  select(department, M_i, pi_i)

```

```

# A tibble: 20 x 3
  department                M_i pi_i
  <chr>                <dbl> <dbl>
1 Biology                77 1
2 Economics              81 1
3 Mathematics            82 1
4 Music                  62 1
5 Physics                60 1
6 Romance Studies        66 1
7 Psychology & Neuroscience 59 1
8 Art, Art History & Visual Studies 44 0.814
9 Asian & Middle Eastern Studies 45 0.832
10 Cinematic Arts         7 0.129
11 Computer Science       48 0.888
12 Dance Program          21 0.388
13 English                50 0.925
14 Gender, Sexuality & Feminist Studies 13 0.240
15 History                47 0.869
16 Philosophy             24 0.444
17 Political Science      43 0.795
18 Sociology              35 0.647
19 Statistical Science     32 0.592
20 Writing and Rhetoric Program 31 0.573

```

```
# stage 2 srs

# srs 5 faculty from each department
pps_sample$faculty_numbers <- lapply(
  pps_sample$M_i,
  function(M_i) sample(1:M_i, 5)
)

# view numbers of faculty selected
pps_sample |>
  select(department, M_i, pi_i, faculty_numbers)
```

```
# A tibble: 20 x 4
  department                M_i  pi_i faculty_numbers
  <chr>                <dbl> <dbl> <list>
1 Biology                77 1    <int [5]>
2 Economics              81 1    <int [5]>
3 Mathematics            82 1    <int [5]>
4 Music                  62 1    <int [5]>
5 Physics                60 1    <int [5]>
6 Romance Studies        66 1    <int [5]>
7 Psychology & Neuroscience 59 1    <int [5]>
8 Art, Art History & Visual Studies 44 0.814 <int [5]>
9 Asian & Middle Eastern Studies 45 0.832 <int [5]>
10 Cinematic Arts         7 0.129 <int [5]>
11 Computer Science       48 0.888 <int [5]>
12 Dance Program          21 0.388 <int [5]>
13 English               50 0.925 <int [5]>
14 Gender, Sexuality & Feminist Studies 13 0.240 <int [5]>
15 History               47 0.869 <int [5]>
16 Philosophy            24 0.444 <int [5]>
17 Political Science      43 0.795 <int [5]>
18 Sociology              35 0.647 <int [5]>
19 Statistical Science     32 0.592 <int [5]>
20 Writing and Rhetoric Program 31 0.573 <int [5]>
```

```
samples <- read.csv("pps_sample.csv")

samples <- samples |>
  select(-any_of(c("M_i", "pi_i", "pi", "weight"))) |>
  rename(
```

```

    woman = Sex...1...F..0...other.,
    grant = Ongoing.grant...1...Yes..0...No.,
    has_phd = PhD...1...Yes..0...No.,
    yrs_phd = Number.of.years.since.Phd,
    yrs_pub = Years.since.last.pub,
    intl_ugrad = International.undergrad.degree.
  )

# 3. attach M_i and pi_i, then build the weight
samples <- samples |>
  left_join(dept_info, by = "department")

samples$weight <- (t_x / (n_dept * samples$M_i)) * (samples$M_i / 5)

# 4. survey design
fac_des <- svydesign(
  ids = ~department,
  weights = ~weight,
  data = samples
)

```

Question 1. What is an estimate of the fraction of faculty who are women?

```

#|label: women-prop

prop_women <- svymean(~woman, fac_des)
prop_women

```

```

      mean      SE
woman 0.37 0.0567

```

```

confint(prop_women, level = .95)

```

```

      2.5 %    97.5 %
woman 0.2588545 0.4811455

```

Question 2. What fraction of faculty have at least one grant that is ongoing in 2026?

```
#|label: grant
```

```
prop_grant <- svymean(~grant, fac_des)  
prop_grant
```

```
      mean      SE  
grant 0.19 0.0552
```

```
confint(prop_grant, level = .95)
```

```
      2.5 %    97.5 %  
grant 0.08180416 0.2981958
```

Question 3. Among faculty who have a Ph.D. degree, what is the average number of years since they received the Ph.D. degree?

```
#|label: phd
```

```
phd_des <- subset(fac_des, has_phd == 1)
```

```
avg_phd <- svymean(~yrs_phd, phd_des)  
avg_phd
```

```
      mean      SE  
yrs_phd 25.703 2.5448
```

```
confint(avg_phd, level = .95)
```

```
      2.5 %    97.5 %  
yrs_phd 20.71508 30.69032
```

Question 4.

a) Among those who have published papers, what is the avg number of years since their last publication?

```
#|label: pub-yrs

pub_des <- subset(fac_des, !is.na(yrs_pub))

avg_pub <- svymean(~yrs_pub, pub_des)
avg_pub
```

```
      mean      SE
yrs_pub 2.8481 0.6691
```

```
confint(avg_pub, level = .95)
```

```
      2.5 %   97.5 %
yrs_pub 1.536777 4.159426
```

b) Did this faculty receive their undergraduate degree outside of the US?

```
#|label: intl-ug

prop_intl <- svymean(~intl_ugrad, fac_des)
prop_intl
```

```
      mean  SE
intl_ugrad  NA NaN
```

```
confint(prop_intl, level = .95)
```

```
      2.5 % 97.5 %
intl_ugrad  NaN   NaN
```